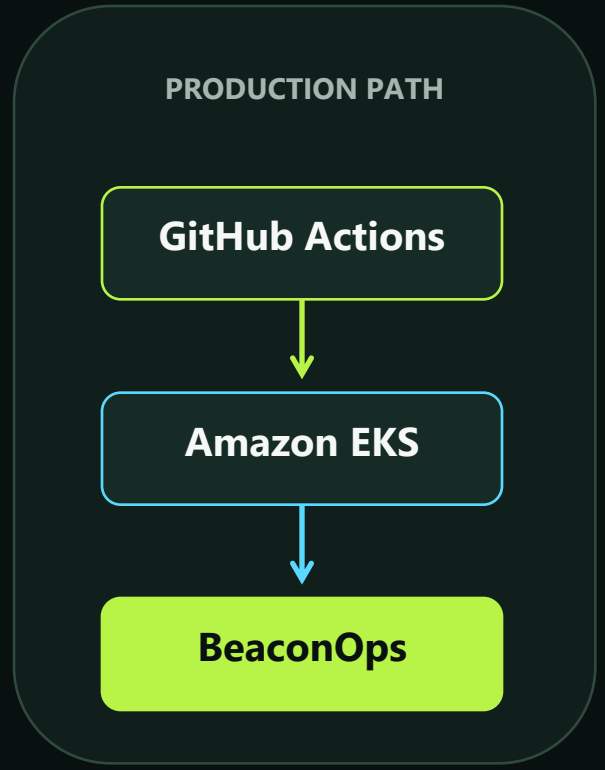
DEVOPS FINAL CAPSTONE

BeaconOps

Production incident command center on Amazon EKS



Kabeer-Scaler

October 2026

github.com/Kabeer-Scaler/beaconops-capstone

02

Project overview

A full stack incident tracker with a production delivery path

Purpose

BeaconOps gives an operations team one place to declare incidents, assign command, track severity and update resolution status.

Incident lifecycle



8

API tests in CI

2 + 2

frontend and backend replicas

1

live EKS production target

SHA

immutable image tags

Source: repository README, Helm production values and successful GitHub Actions run #9

03

Technology stack

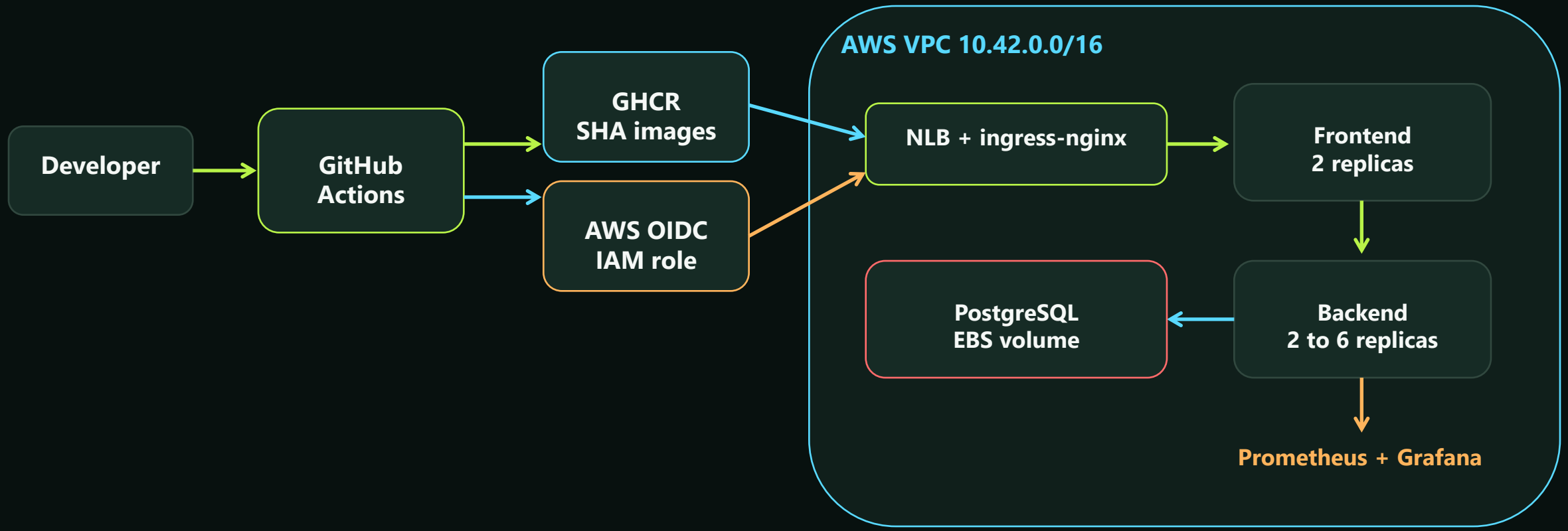
Each layer supports a specific part of the production path

APPLICATION	React + Vite frontend FastAPI + SQLAlchemy backend
PACKAGING	Docker multi-stage builds GitHub Container Registry
PLATFORM	Amazon EKS Helm 3 ingress-nginx metrics-server
DATA	PostgreSQL 16 Amazon EBS CSI 5 Gi persistent volume
DELIVERY + OPS	GitHub Actions Terraform Trivy Prometheus Grafana

Source: application manifests, Dockerfiles, Terraform configuration and CI workflow

Production architecture

Source changes travel through a secure delivery path into a private worker tier



Source: Terraform VPC and EKS modules, GitHub Actions workflow and Helm chart

05

CI/CD flow

Three gated jobs move a commit from source validation to an atomic release

1 TEST SOURCE

Python 3.12
pytest: 8 tests
compileall
Node.js 22
npm ci + npm build

2 BUILD, SCAN, PUBLISH

Buildx images
Trivy HIGH and CRITICAL gate
GHCR login
Push frontend and backend
Tag with commit SHA

3 DEPLOY TO EKS

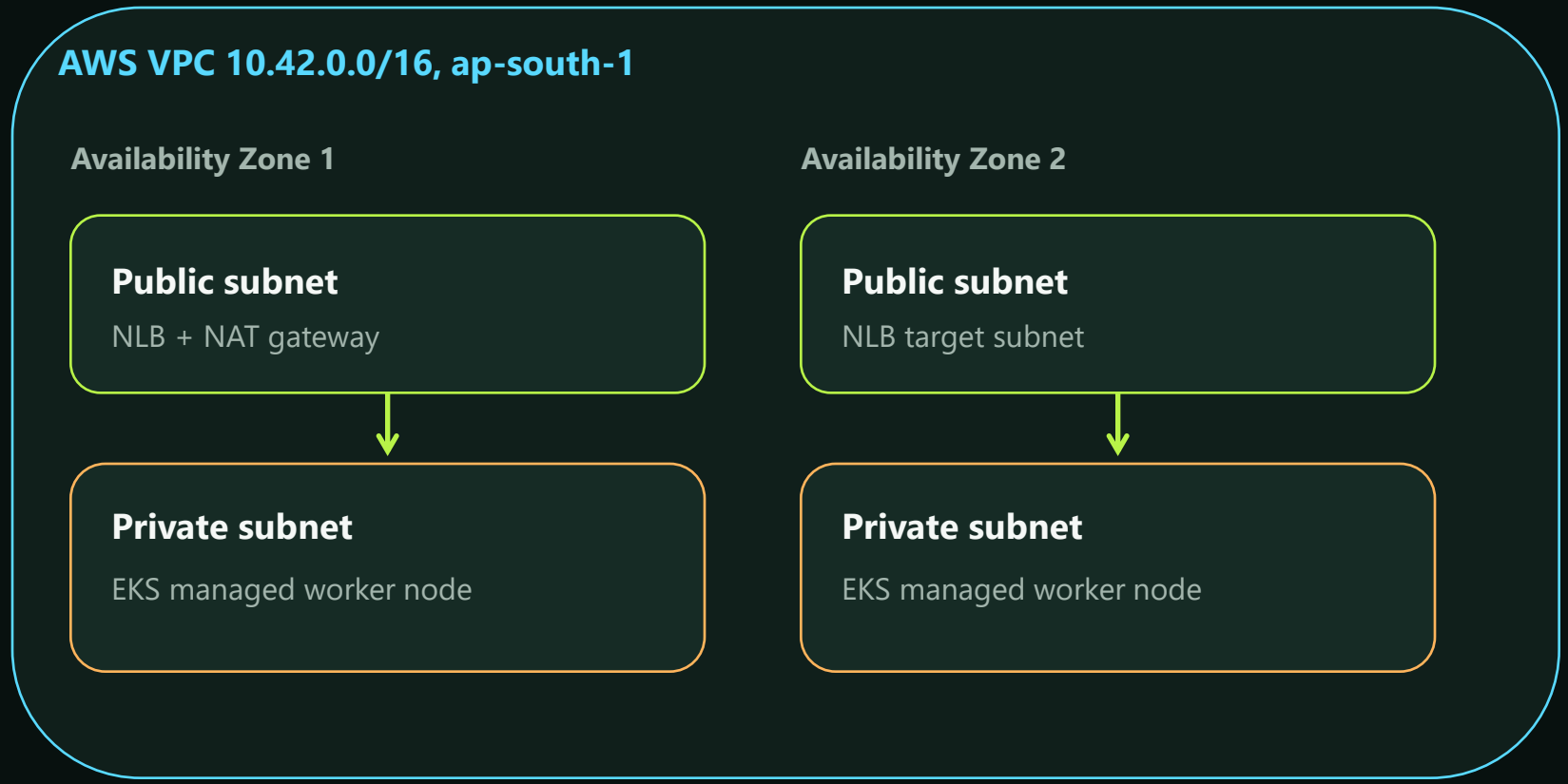
GitHub OIDC to AWS
Install platform charts
Helm atomic upgrade
Wait for rollouts
Publish NLB endpoint

Verified production run #9, commit 6a1f963, all three jobs passed

Source: .github/workflows/ci-cd.yml and GitHub Actions run 38919476385

Terraform infrastructure

A two-AZ VPC keeps worker nodes private while the load balancer remains public



CONTROL + IDENTITY

EKS 1.37

Managed control plane with public API endpoint and creator admin access.

Worker group

t3.medium, desired 2, minimum 2, maximum 4, 30 Gi disks.

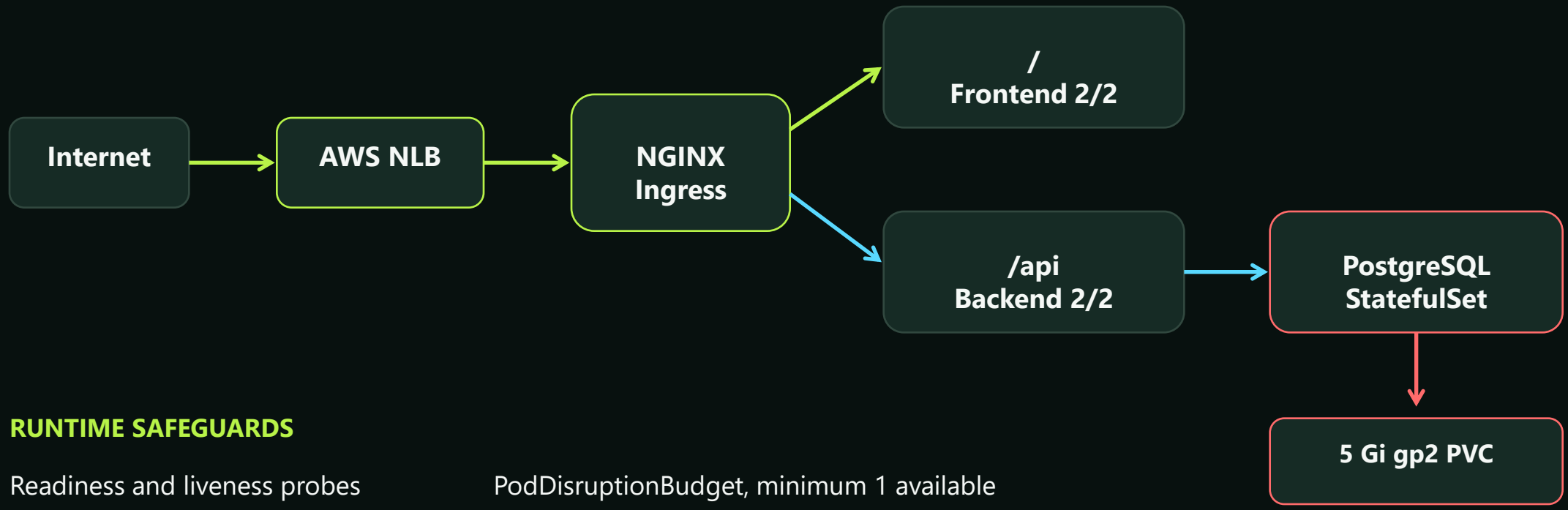
Identity + storage

GitHub OIDC deploy role and EBS CSI Pod Identity association.

Source: terraform/main.tf and terraform/variables.tf

Kubernetes setup

Helm defines routing, scaling, persistence and workload safeguards



RUNTIME SAFEGUARDS

Readiness and liveness probes

PodDisruptionBudget, minimum 1 available

Backend HPA, 2 to 6 replicas at 60% CPU

Source: Helm production values and Kubernetes templates

08

Observability

Application metrics connect deployment health to actionable dashboards



DASHBOARD PANELS

HTTP request rate

p95 request latency

5xx errors over 5 minutes

Ready backend pods

Backend CPU usage

Grafana refresh: 15 seconds

Prometheus Operator discovers the ServiceMonitor by release label.

Source: ServiceMonitor and Grafana dashboard templates in helm/beaconops/templates

09

Security and reliability

Controls span identity, image supply chain, runtime isolation and rollback

Short-lived AWS access

GitHub OIDC assumes a repository and environment scoped IAM role. CI stores no long-lived AWS access keys.

Container security gate

Trivy blocks HIGH or CRITICAL findings before any image reaches GHCR or the cluster.

Hardened workloads

Containers run without root, drop Linux capabilities and use read-only filesystems where supported.

Safe releases

Helm uses atomic upgrades and rollout waits. Failure diagnostics capture pods, PVCs, logs and events.

Immutable SHA tags tie every running image to a reviewed Git commit.

Source: GitHub Actions workflow, Dockerfiles, Terraform IAM policy and Helm templates

Key learnings and outcome

The final system works because delivery details received the same attention as application code

01	**OIDC trust must match modern GitHub subjects**	Immutable owner and repository IDs fixed AssumeRoleWithWebIdentity failures.
02	**Persistent storage needs identity and topology**	EBS CSI Pod Identity plus an explicit gp2 StorageClass resolved provisioning delays.
03	**Container users need matching filesystem ownership**	Runtime UID and directory permissions removed startup failures in hardened images.
04	**Atomic deploys still need diagnostics**	Background pod, PVC, log and event capture made timeout failures actionable.

Outcome: a tested, scanned, reproducible application deployed successfully to Amazon EKS